

ANSWER KEY — Integration Lesson 33 Test: Improper Integrals

For the grader · full working shown · full credit needs the limit written out, not ∞ plugged in as a number

1. Find each limit: (a) $\lim_{b \rightarrow \infty} 1/b^2$ (b) $\lim_{b \rightarrow \infty} (4 + 1/b)$ (c) $\lim_{b \rightarrow \infty} \ln b$

Work: (a) $1/b^2$ shrinks to nothing as b grows ($1/100, 1/10000, \dots$) $\rightarrow 0$. (b) the $1/b$ piece fades to 0, leaving the $4 \rightarrow 4$. (c) $\ln b$ climbs slowly but never stops $\rightarrow \infty$ (no finite limit). **Answer: (a) 0 (b) 4 (c) ∞ .**

2. Evaluate $\int_1^\infty 1/x^5 dx$.

Work: antiderivative of x^{-5} is $x^{-4}/(-4) = -1/(4x^4)$. So $\int_1^b = [-1/(4x^4)]_1^b = -1/(4b^4) + 1/4$. As $b \rightarrow \infty$, $1/(4b^4) \rightarrow 0$, so the limit is $1/4$. (p-rule check: $p = 5 > 1$, value $1/(5-1) = 1/4$ ✓.) **Answer: converges to 1/4.**

3. Evaluate $\int_3^\infty 1/x^2 dx$.

Work: $\int_3^b 1/x^2 dx = [-1/x]_3^b = -1/b + 1/3$. As $b \rightarrow \infty$, $-1/b \rightarrow 0$, leaving $1/3$. (The p-rule value $1/(p-1) = 1$ is only for a lower limit of 1 — starting at 3 skips the fattest part of the tail.) **Answer: converges to 1/3.**

4. Show that $\int_1^\infty 1/x^{1/3} dx$ diverges, and explain why.

Work: $\int x^{-1/3} dx = x^{2/3}/(2/3) = (3/2)x^{2/3}$. So $\int_1^b = (3/2)b^{2/3} - 3/2 \rightarrow \infty$ as $b \rightarrow \infty$. Why: shrinking to 0 isn't enough — the tail must shrink FAST ENOUGH, and $1/x^{1/3}$ shrinks even more slowly than the borderline $1/x$ ($p = 1/3 \leq 1$). **Answer: diverges — the tail is too fat.**

5. Evaluate $\int_0^\infty e^{-3x} dx$.

Work: antiderivative of e^{-3x} is $e^{-3x}/(-3) = -1/3e^{-3x}$. So $\int_0^b = [-1/3e^{-3x}]_0^b = -1/3e^{-3b} + 1/3$. As $b \rightarrow \infty$, $e^{-3b} \rightarrow 0$, leaving $1/3$. **Answer: converges to 1/3.**

6. p-rule: (a) $\int_1^\infty 1/x^6 dx$ (b) $\int_1^\infty 1/x^{0.8} dx$

Work: the p-rule: $\int_1^\infty 1/x^p dx$ converges exactly when $p > 1$, with value $1/(p-1)$. (a) $p = 6 > 1 \rightarrow$ converges to $1/(6-1) = 1/5$. (b) $p = 0.8 \leq 1 \rightarrow$ diverges. **Answer: (a) converges to 1/5 (b) diverges.**

7. Evaluate $\int_0^9 1/\sqrt{x} dx$.

Work: spike at $x = 0$, so use a wall at a : $\int_a^9 x^{-1/2} dx = [2\sqrt{x}]_a^9 = 2 \cdot 3 - 2\sqrt{a} = 6 - 2\sqrt{a}$. As $a \rightarrow 0^+$, $\sqrt{a} \rightarrow 0$, so the limit is 6. An infinitely tall spike with area exactly 6. **Answer: converges to 6.**

8. Show that $\int_0^1 1/x \, dx$ diverges.

Work: $\int_a^1 1/x \, dx = [\ln x]_a^1 = \ln 1 - \ln a = -\ln a$. As $a \rightarrow 0^+$, $\ln a \rightarrow -\infty$, so $-\ln a \rightarrow +\infty$ (check: $a = 0.001$ gives $-\ln 0.001 \approx 6.9$; $a = 10^{-9}$ gives ≈ 20.7 — it keeps climbing). The area never settles.

Answer: diverges — the spike at 0 is too fat.

9. What does CONVERGE mean, and how can an infinitely long region have finite area?

Work (sample answer): converge means: as the wall b slides out to infinity, the areas $\int_1^b$ head toward one fixed finite number — that number is the area. It's possible because the curve's tail shrinks so fast that each new stretch adds almost nothing: for $1/x^2$, the area up to b is $1 - 1/b$ (0.9, then 0.99, then 0.999...), settling at exactly 1. Like $1/2 + 1/4 + 1/8 + \dots = 1$, infinitely many pieces can total a finite amount. **Give credit for: "the running area settles toward one number because the added slivers shrink fast enough."**

10. Challenge: evaluate $\int_2^\infty 1/(x(x-1)) \, dx$.

Work: partial fractions: $1/(x(x-1)) = A/x + B/(x-1) \rightarrow 1 = A(x-1) + Bx$. At $x = 0$: $A = -1$; at $x = 1$: $B = 1$. So the integrand is $1/(x-1) - 1/x$. Then $\int_2^b = [\ln(x-1) - \ln x]_2^b = [\ln((x-1)/x)]_2^b = \ln((b-1)/b) - \ln(1/2)$. As $b \rightarrow \infty$, $(b-1)/b \rightarrow 1$, so $\ln((b-1)/b) \rightarrow \ln 1 = 0$. Total: $0 - \ln(1/2) = \ln 2 \approx 0.693$. **Answer: converges to $\ln 2$.**